# An Introduction to 3D Printing

## Part 3: Science

*Vibrations and Waves, Linear Motion, and Thermal Expansion*

# Outline

**Part 1: Printers**

Process

Limitations

Advantages

**Part 2: Practical Application**

Examples

Critical Thinking

Design

**Part 3: Science**

Vibrations and Waves

Linear Motion

Thermal Expansion

# Vibrations and Waves

- Tensioned belts (and pulleys and bearings) connect motors and the toolhead.
- The printer creates vibrations when the toolhead:
  - Speeds up.
  - Slows down.
  - Changes direction.
- Belts (strings) + vibrations (waves) = Physics.
  - Motions systems with belts have a natural frequency.
  - What does this mean?
  - *Hint:* Think of a guitar or piano string.

# Repeating Mistakes

- What happens when the toolhead hits a natural frequency?
    - Resonance.
- Toolhead excites the belts (adds energy) with vibrations.
- Resonance maximizes vibrations.
    - Standing waves undergo constructive interference.

# Repeating Mistakes

- Vibrations make the belts push and pull on the toolhead.
- This creates small errors in nozzle position.
- **Errors repeat based on the frequency.**
- Imagine trying to write on paper with someone bumping your arm at a constant time interval.

- Where are the repeating errors?
- Sharp corners.
- Which direction was the toolhead moving?
- From left to right.

(Screenshot from Teaching Tech.)

- How can you determine frequency?
- Lookup printing speed in settings.
- Measure distance between repeating patterns.

$$freq = \frac{print\_speed}{repeating\_dist}$$

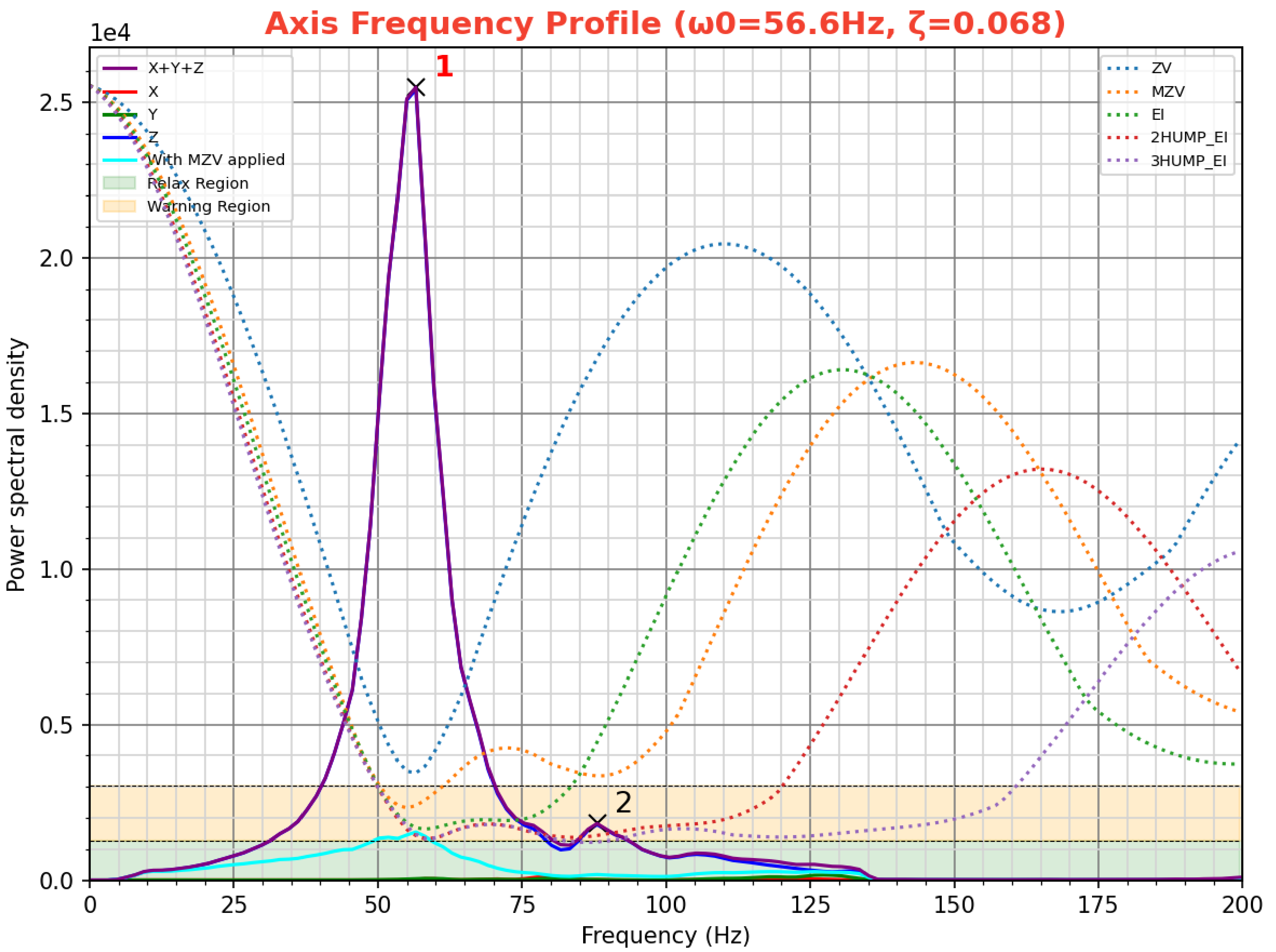
(Screenshot from Teaching Tech.)

# Input Shaping

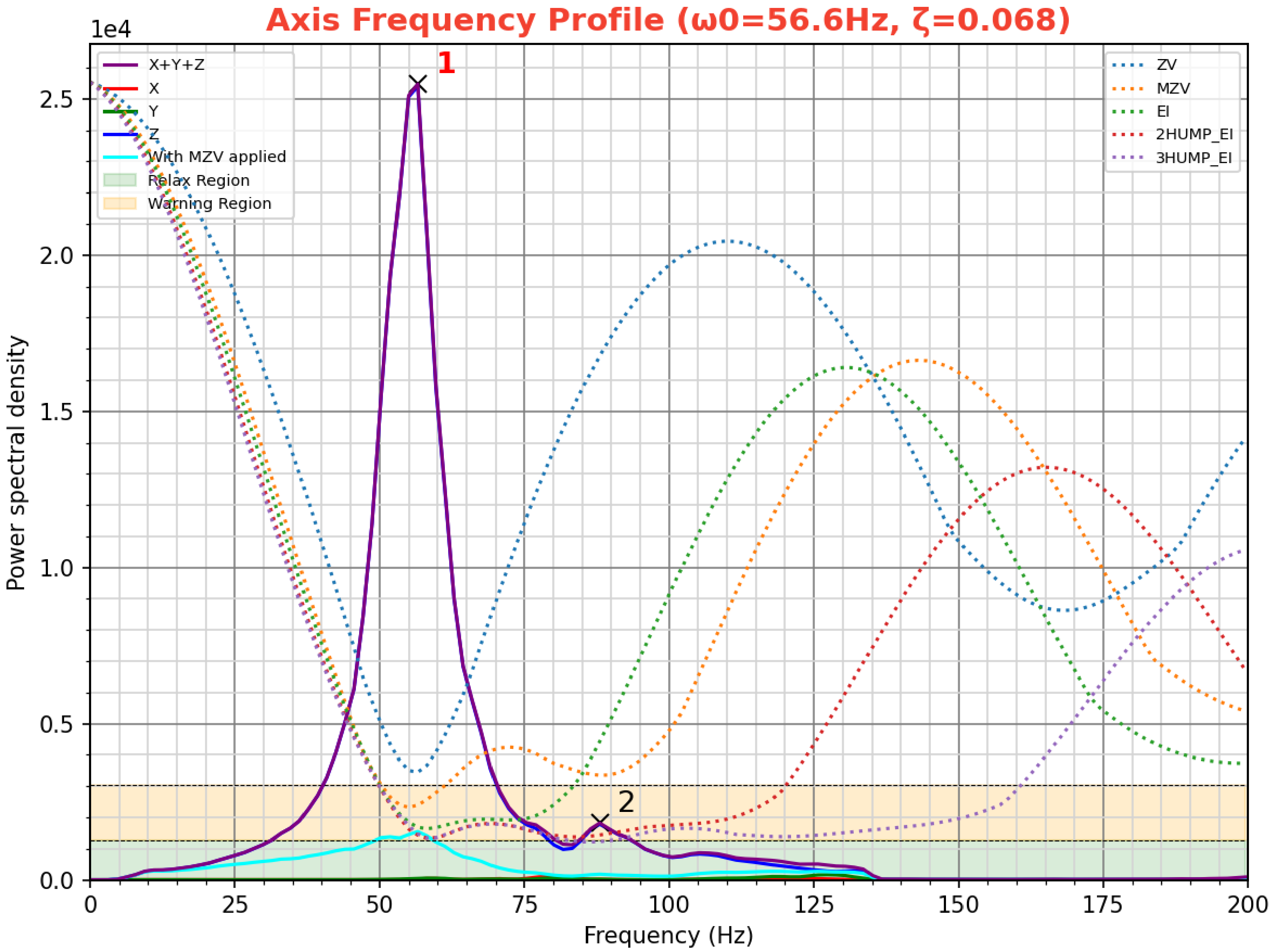
- How can you fix vibrations?
- Input shaping adds extra movements that cancel vibrations.
  - Destructive wave interference.
- Similar to noise-canceling headphones.

(Demo by ACS Motion Control.)

- Shake and Tune Plugin.
- Attach an accelerometer to the toolhead.
- Move the toolhead back and forth to sweep frequencies.
- Script analyzes the results, makes this graph, and suggests Input Shaper settings.



- X axis results for my printer.
- Single and narrow peak is ideal.
- **Purple** → Before
- **Cyan** → After
- Ignore high frequencies on the right side of the graph.
- Each printer and each axis has a characteristic shape.



# Speed Limits

I want to print faster.  Here is what I have:

- *Acceleration*: 10,000 $mm/s^2$
- *Max speed:* 300 mm/s
- *Nozzle diameter:* 0.4 mm
- *Layer height:* 0.2 mm
- *Max volumetric flow:* 22 $mm^3/s$
- Should I upgrade motors or hot end?
- What about top speed or acceleration?

# Velocity and Acceleration

- What is the distance to reach top speed?
  - Starting speed (cornering speed, jerk): 5 mm/s
  - Increasing cornering speed *greatly* increases vibrations.
- What is the smallest extrusion that reaches top speed?
  - 9 mm to accelerate *and* decelerate.

$$v^2 = {v_0}^2 + 2a(x - x_0)$$

$$300^2 = 5^2 + 2 \times 10,000(dist)$$

$$dist = 4.5\ mm$$

# Velocity and Flow

Can my hotend melt enough plastic at top speed?

- Almost.

$$print\_speed = \frac{flow\_rate}{cross\_section}$$

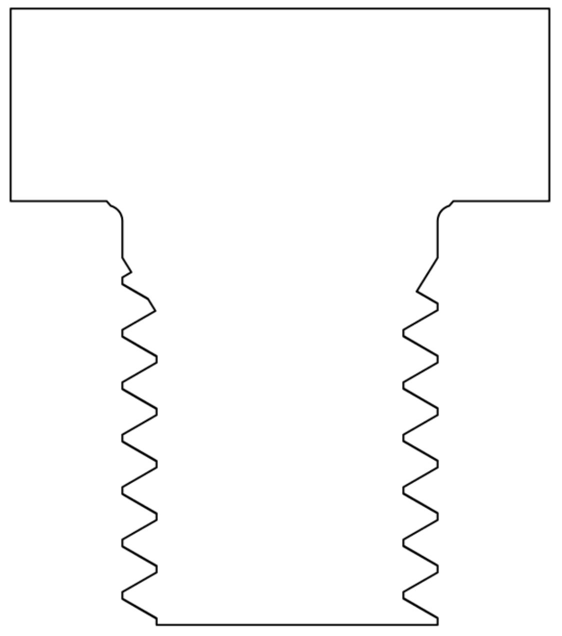
$$print\_speed = \frac{flow\_rate}{extrusion\_width \times layer\_height}$$

$$print\_speed = \frac{22}{0.4 \times 0.2}$$

$$print\_speed = 275\ mm/s$$

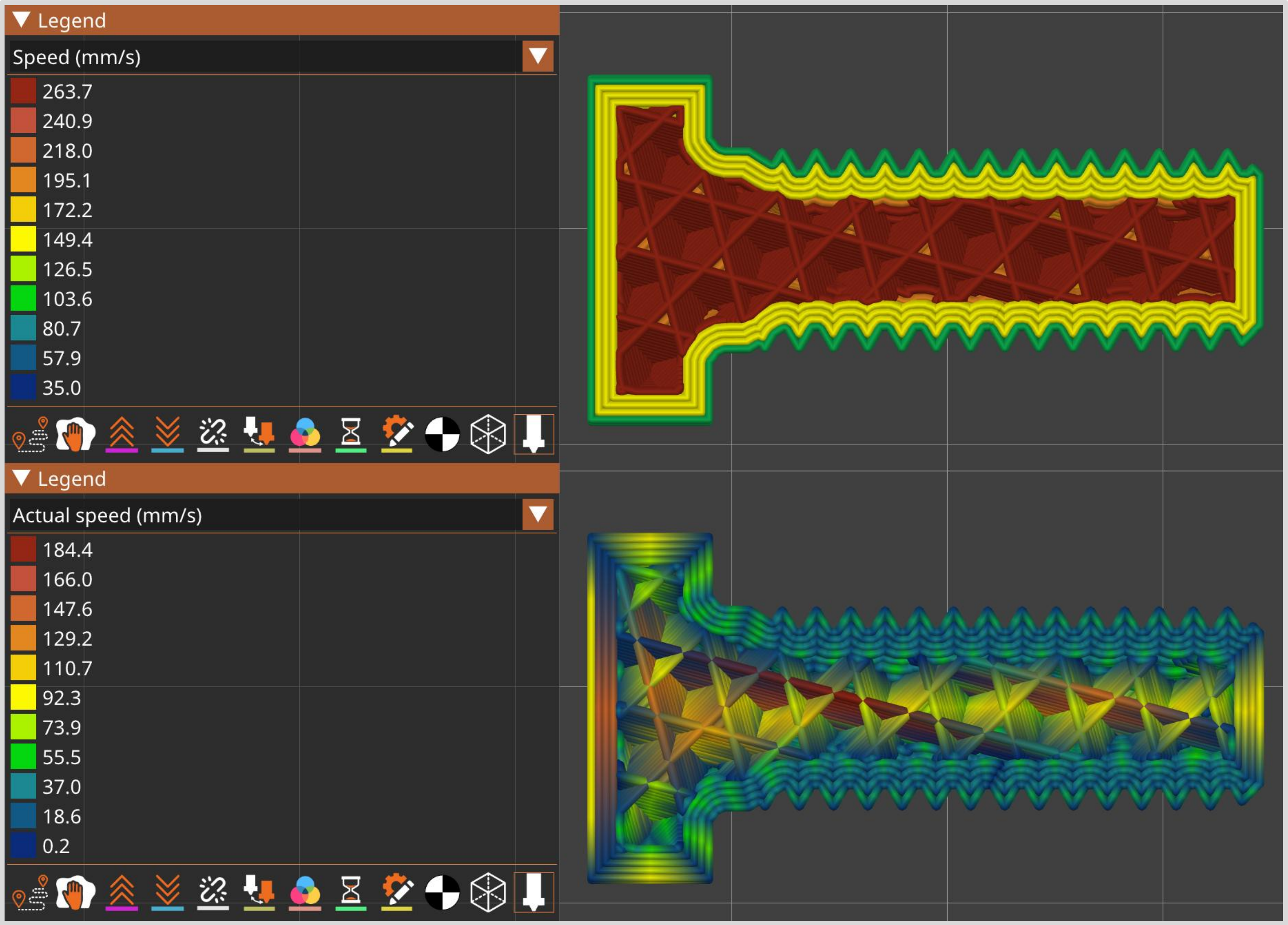
# Finding Limits

- Real life limits are more complex and nuanced.
- *Example:* Printed bolt
  - 32 mm long (about 1 inch)
  - 8 mm diameter (about 3/8 inch).
- What is going to limit my printer?
- Will the printer hit top speed (about 275 mm/s)?
  - The dimensions are under 9mm …
- Let PrusaSlicer do all the math!



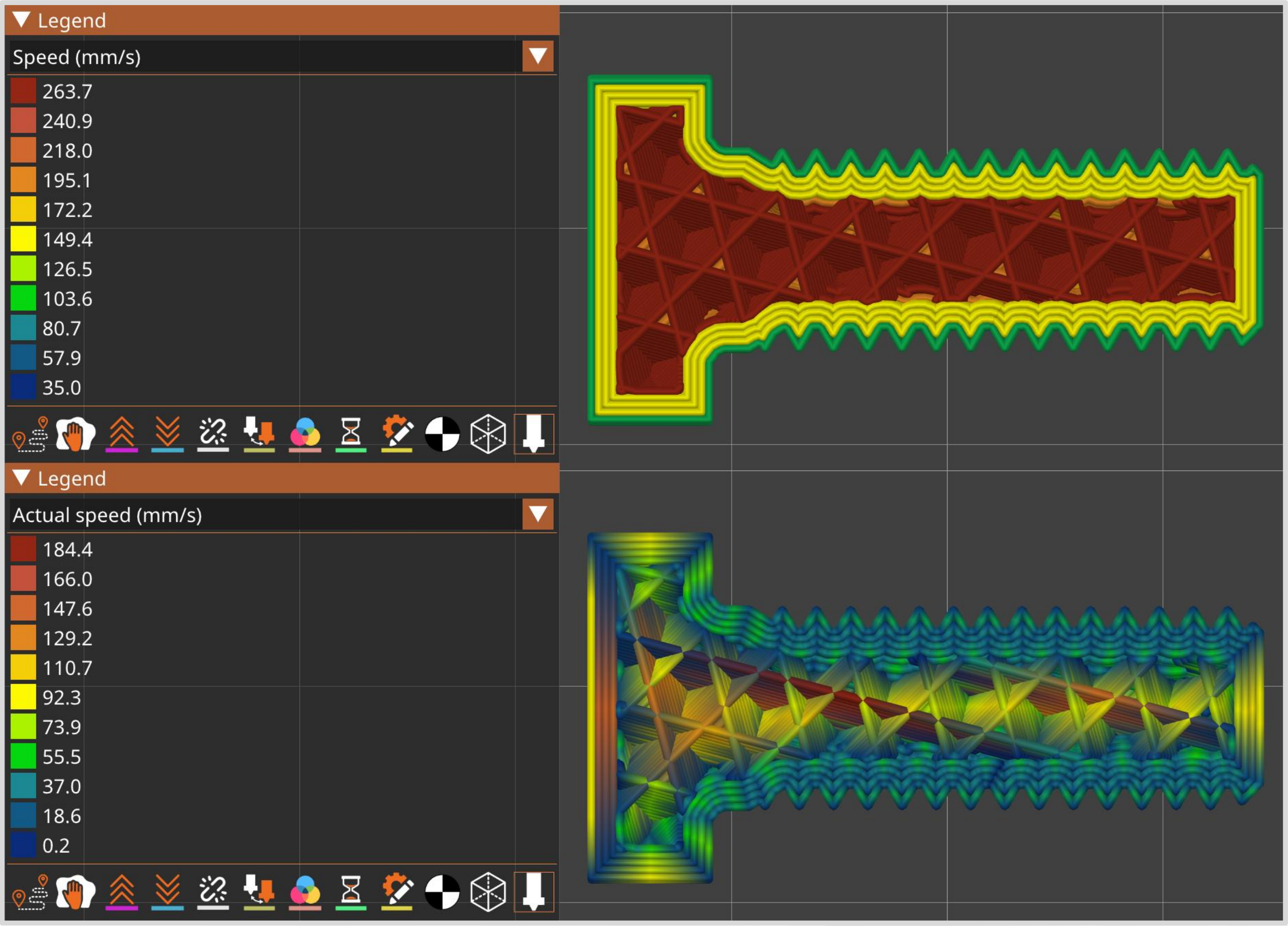
- Max requested speed: **264 mm/s**

- Max actual speed: **184 mm/s**

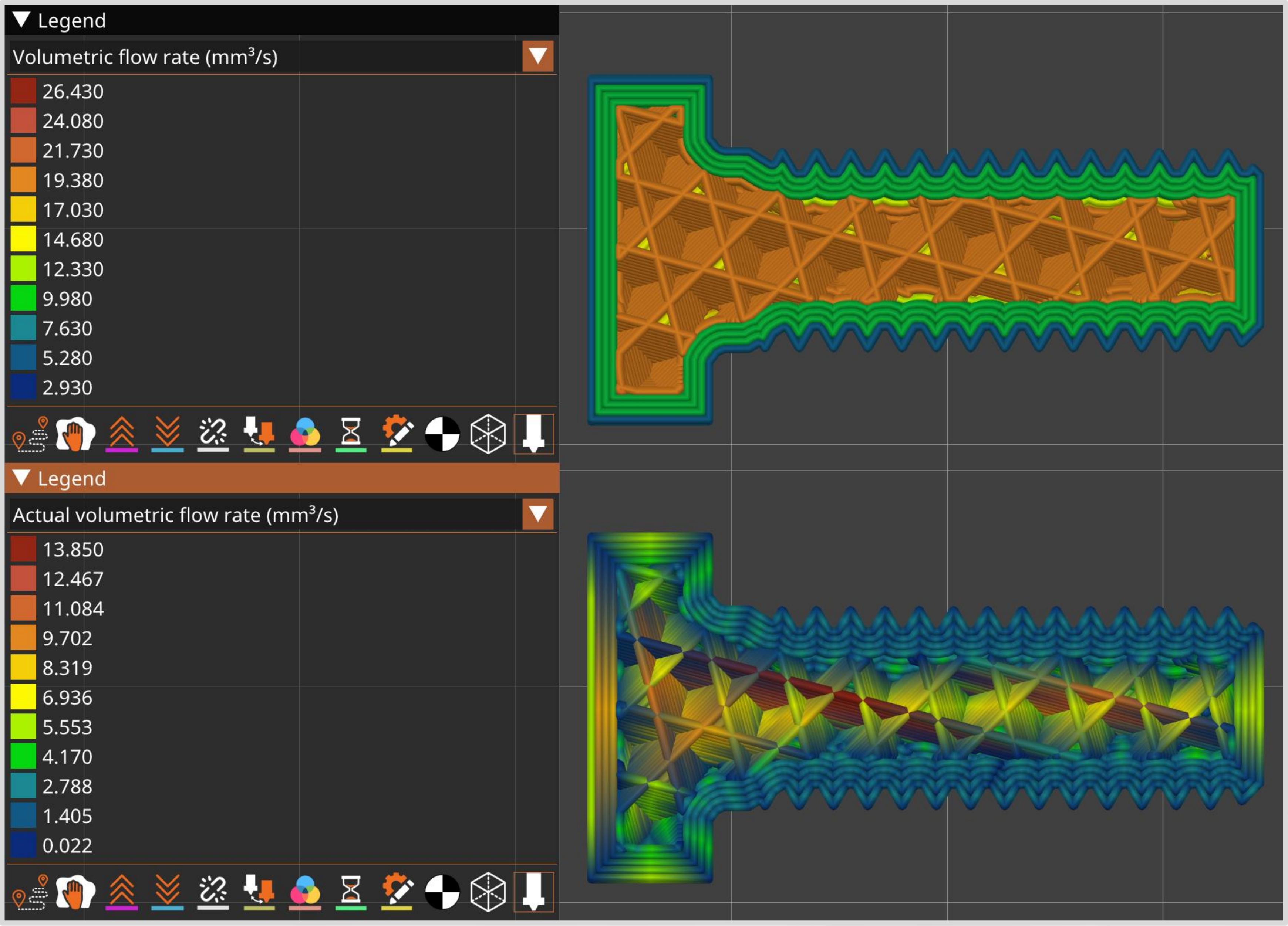


Why is the requested speed non-uniform?

- Adjust speeds according to geometry.
- Slow and pretty on the outside.
- Fast and rough on the inside.



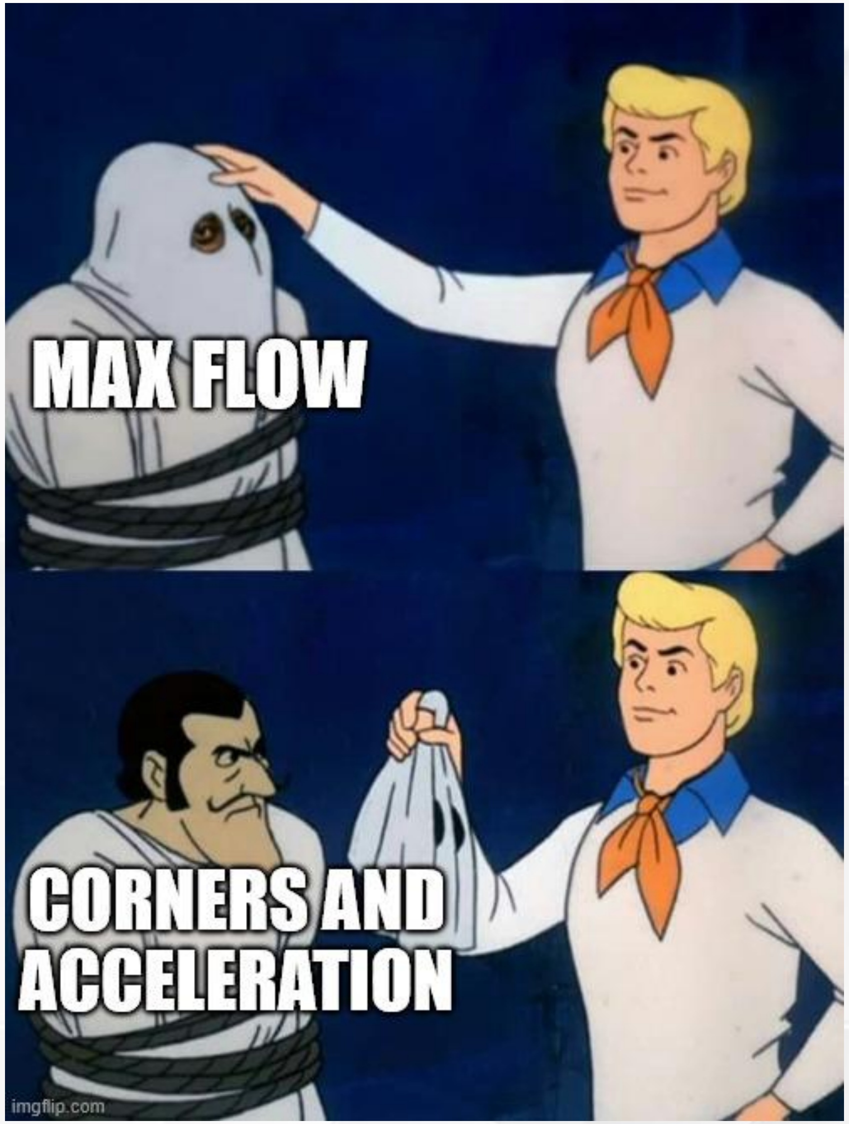
- Max requested flow: **22 mm³/s**




- Max actual flow: **14 mm/s**

# Speed Limits

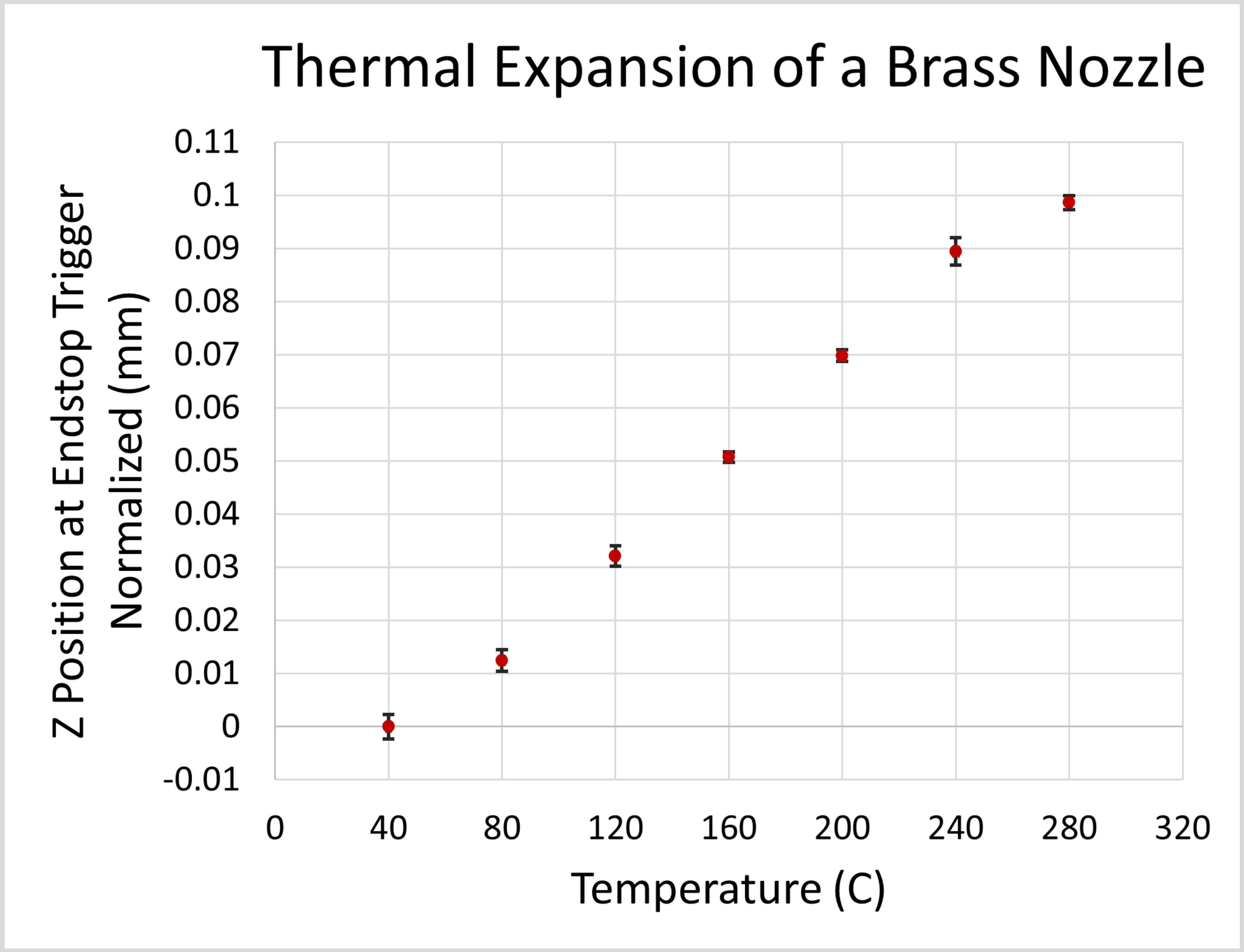
- Max requested speed: 264 mm/s
  - Close to estimate for flow-limited top speed.
- Max actual speed: 184 mm/s
- Max requested flow: 22 $mm^3/s$
- Max actual flow: 14 $mm^3/s$
- My printer does not hit max speed, or max flow.
- Limited by acceleration and geometry (corners).



# Thermal Expansion

- Metal expands as it heats.
  - Coefficient of linear expansion.
- Estimate brass nozzle using a 12mm cylinder:
  - $\Delta T = 150°C \rightarrow +0.034$ mm.
  - $\Delta T = 300$ °C $\rightarrow +0.068$ mm.
  - Does not include effects from heatsink or heat break.
- Is this large enough to affect the first layer?
  - **Yes, first layers are very sensitive.**
  - Well known from experience.

- Data from my own printer.
- Set temperature and wait 10 minutes.
- 19 samples per point.
- My printer uses two probes and an offset calculation → normalized to zero.
- Error bars ± one standard deviation.



# Thermal Expansion

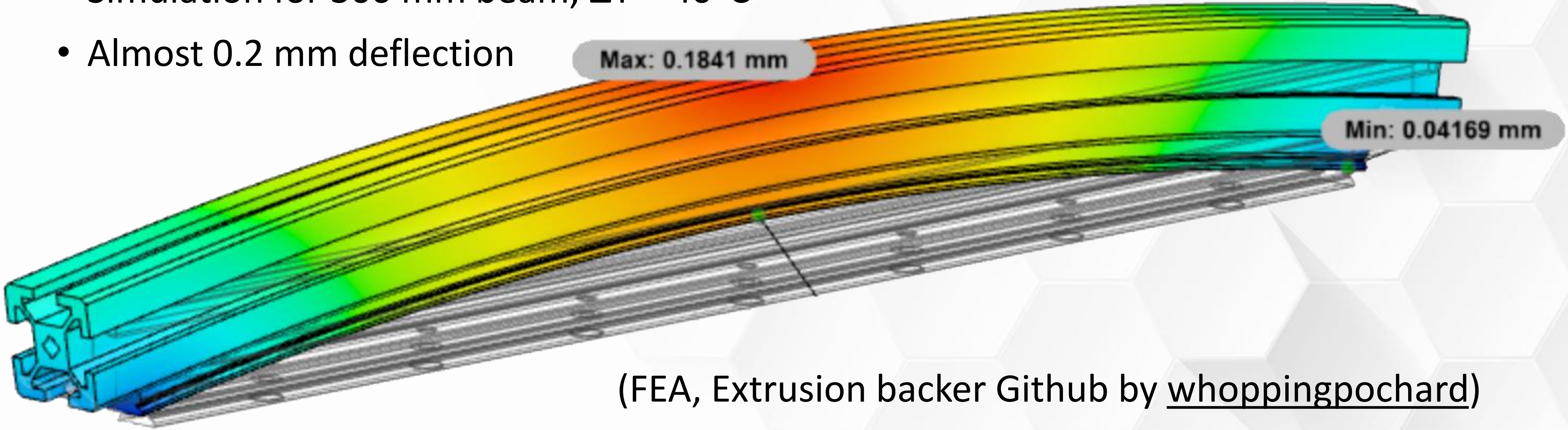
- What can you do about it?
  - “Heat soak.”  Let everything expand and wait for equilibrium.
  - Then probe nozzle position.
  - First layer is over-squished if you skip waiting.
  - Print beds need this same treatment.

# Thermal Expansion

- Printers have lots of parts.
- What happens when you assemble metals with different expansion rates?
  - Frame extusions
  - Sheet metal
  - Screws
  - Bearings
  - Rods and rails

# (Bimetallic) Thermal Expansion

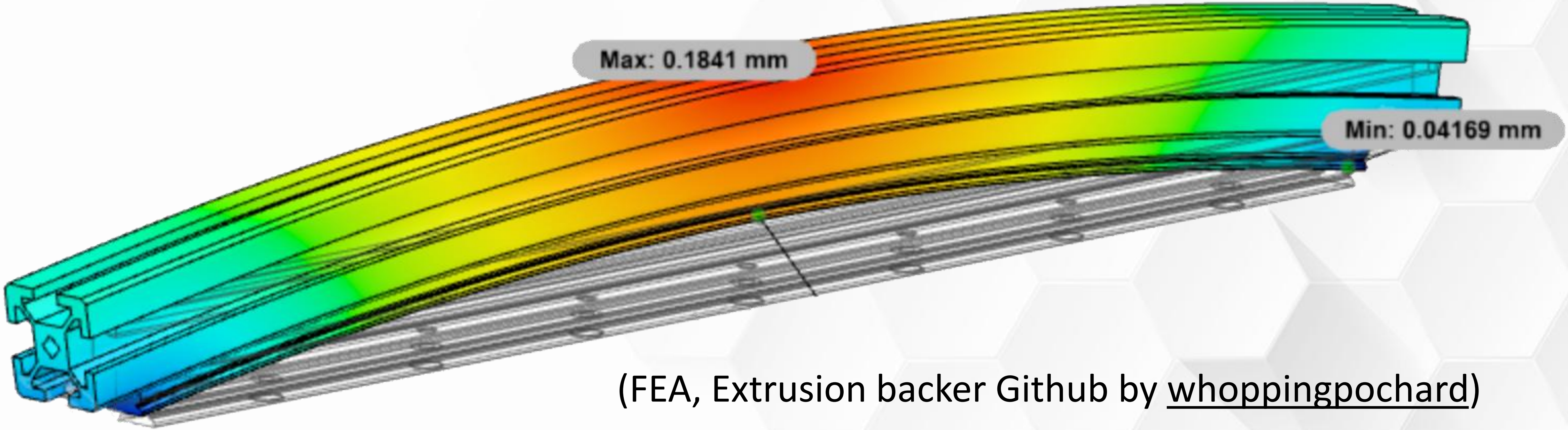
- Aluminum extrusion on top.
- Stainless-steel linear rail on bottom.
- Bimetallic “strips” **bend** and form an arc.
- Simulation for 300 mm beam, $\Delta T = 40^\circ C$
- Almost 0.2 mm deflection




(FEA, Extrusion backer Github by whoppingpochard)

# (Bimetallic) Thermal Expansion

- Old fashioned thermometers used bimetallic strips to make the dial.
- Which material has higher expansion coefficient?
- Aluminum because it is on the outside of the curve.




(FEA, Extrusion backer Github by whoppingpochard)


THANK YOU
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